

OBJECTIVE

Looking for more opportunities in **Artificial Intelligence** with **Computer Vision** / **NLP** / **Reinforcement Learning** after having about 14 years of overall experience in the field

EDUCATION

Regents University London September 2016 – September 2017

M.A. in International Business (Chartered Management Institute Qualification – Level 7 Extended Diploma)

Dissertation Title: ***How can luxury car brand manufacturers compete against the IT giants on their in-car infotainment systems?*** (uses machine learning algorithms such as Collaborative Filtering and Analytical Hierarchical Processing)

Electives: Private Equity and Venture Capitals, Business Analytics (Certificate in SAS JMP)

University of Southern California September 2007 – May 2009

MS in Electrical Engineering

Course work: Image/Video/Speech Processing, Digital/Analog Circuits and RF Systems

Birla Institute of Technology and Science August 2003 – May 2007

B.E. (Hons) Electrical and Electronics Engineering

Focus on Electrical and Electronics.

CERTIFICATES

Huggingface February 2025

Certificate in **Deep Reinforcement Learning**

In Progress: Focus on Stable-Baselines and PPO in Colab

Deeplearning.ai May 2021

Certificate in Generative AI with Large Language Models

Focus on Amazon SageMaker

Deeplearning.ai May 2021

Certificate in two of three courses in the Generative Adversarial Networks Specialization

Focus on Tensorflow

Deeplearning.ai July 2020

Certificate in the four course Natural Language Processing Specialization

Focus on Trax

Persontyle April 2018

Certified AI Business Professional (CAIBP)

Focus on Use Case Definition, KPIs, Big Data System Development and Machine Learning

Deeplearning.ai February 2018

Certificate in the five course Deep Learning Specialization

Focus on Python, Tensor Flow and Keras

Simplilearn July 2014

Certificate in Big Data Technologies

Focus on Hadoop and MapReduce Framework

EMPLOYMENT

Nayan India Science & Technologies Pvt. Ltd., India.....(Apr. 2024 – Present)

Computer Vision Engineer

- Designed **real-time CCTV video parser** that parses 10 second videos upon detection of a vehicle with readable license plates using **tiny YOLOv4** models
- Designed **real-time ANPR system to track and upload vehicles** with license plates and **OCR** onto a dashboard from CCTV footage
- Managed the upgradation of **multi-class non-functional traffic and infrastructure detection, helmet violation, spitting violation and uncovered truck debris** workflows and **MongoDB** operations
- Designed a workflow scaler to achieve real-time performance on detections, annotations and events creation on the dashboard

Giggso Pvt. Ltd., India.....(Sep. 2023 – Jan. 2024)

Senior AI/ML Engineer

- Designed Root-Cause-Analysis for Logitech Bug reports using **DBSCAN clustering** algorithm and **NLP semantic models**
- Performed **data cleaning using CleanLab AI** and trained a **Roberta model** to classify product titles into their respective categories on client's website. **Achieved 97% accuracy** in classification
- Designed and managed the development of **Email Automation system** for shared inbox using **Mistral LLM**

Lokibots Inc., India.....(Aug. 2021 – Jul. 2023)

Global AI/ML Lead

- Managed **Doc AI** projects that employed Huggingface vision models and Donut model in **Python**
- Developed **NLP based Conversational AI** framework for answering questions from pdf documents using Langchain, Llama-Index and FastAPI employing **OpenAI GPT-3** and **Anthropic** in **Python** and deployed in **AWS and GCP platforms**
- Researched on chain-of-thought using cloud-based **LLMs** as well as Huggingface **LLMs**

Acharya Institutes, India.....(May. 2021 – Aug. 2021)

Assistant Professor (Dept. of Computer Science & Engineering)

- Taught undergraduate students on courses in Big Data Analytics and C Programming.
- Mentored and guided graduate students in **AI and ML** projects in **Python**.

Sri Sathya Sai Institute of Higher Learning, India.....(Nov. 2018 – Jun. 2020)

Assistant Professor (Dept. of Mathematics and Computer Science)

- Taught graduate students on courses in **AI and Data Science** such as Soft Computing, Linear Regression, **Machine Learning**, Data Visualization, Deep Learning, **Reinforcement Learning** and Python, and mentored them in their projects (Respiratory Disease Classification, Lung Cancer Detection, **NLP** Feature Extraction from Medical Records, Music Recommendation System).
- Taught undergraduate students on a course in System Software and mentored them in their projects (Online Chess and Optical Character Recognition) in **Python**.
- Collaborated with the Management department on a publication.

Brainpool AI, UK.....(Jan. 2018 – April 2021)

Data Scientist

- Developed and Implemented **AI and Machine Learning** platforms and algorithms for client companies in manufacturing (faulty product detection), **NLP** data processing (non -structured to structured) and communications (data compression) in **Python**.

Garuda Technologies, India.....(Feb. 2014 – Sep.2016)

Founder and Developer

- Research and Development of **Artificial Intelligence** technologies such as Neural Networks based on Deep Learning and **Reinforcement Learning** in **Python**.
- Designed, developed and maintained proprietary Database Software for clients.

Idaho National Laboratory (INL), USA.....(Jan. 2011 – Dec. 2013)

Research Scientist (Machine Learning + Hardware/Software Development)

- *Researched and developed novel extensions to **machine learning** algorithms for green building technology, smart grids, robotics and smart device applications in C++.*
- *Developed a simulation tool of power grids with **Fuzzy Logic** based decision frameworks in C++.*

National Institute for Advanced Transportation Technology (NIATT), USA..(Jan.2010 – Dec. 2010)

Research Scientist (Hardware Design)

- *Delivered the back-end design of the FPGA based SDLC interface hardware for Pre-emptive Traffic Control Systems.*
- *Mentored in University based lab courses on Microcontrollers.*

HACKATHONS

Drug Discovery Hackathon 2020 Phase-I (India) Aug. 2020 – Nov. 2020

- *Lead a team of master's students in developing a Neural Network Regression model to predict the inhibitory activity of candidate molecules with 3CLpro. **Achieved an extremely high-performance prediction model that was highly commended by the panel.***
- *Developed a solution setup using PAFNUCY with the help of team members for applying Reinforcement Learning in generating various ligand poses at several protein binding active sites. **Proposed solution was selected for presentation to the panel.***

SKILLS

Language Skills

- English
- Tamil

IT Skills

- Operating Systems: Windows, Ubuntu
- Cloud: AWS – Step and Lambda functions, Github, Colab, GCP
- Programming: Python, C/C++
- Packages: Langchain, Llama-Index, FastAPI, OpenCV, Tensorflow, Keras, Gym, Stable-Baselines
- Databases: Neo4j, MySQL, MongoDB

TALKS

Institute for Chartered Accountants of India – Muscat (Oman) Chapter May 2018

Talk on “AI in Finance”

Covered details of underlying technologies and its applications in the Finance Sector.

Sohar Hospital – Sohar (Oman) May 2019

Talk on “AI in Medical”

Covered details of underlying technologies and its applications in the Medical Sector.

PUBLICATIONS

Machine Learning / Artificial Intelligence

Book Chapter

- M.S.S. Vinod, U. Ravishankar, N. Sivakumar, “Greening the supply chain through CSR: An assessment of sustainability in NIFTY indexed companies in India using fuzzy based PPP approach”, Sustainable Supply Chains: Strategies, Issues and Models, pp. 133 – 157, 2020.

Journal

- U. Ravishankar, “A Lyapunov based Adaptive and Stable Neural Network Weight Regularization Algorithm”, Neural, Parallel and Scientific Computations (NPSC), vol. 23, pp. 343 – 356, 2015.

Conference

- U. Ravishankar, “Large Language Models for Chain-of-Thought based Automation,” Int’l. Conf. on Recent Advancements in Engineering and Computer Application (ICRAECA), 2023. (Ahmedabad, India)
- U. Ravishankar, “Feasibility study on the Intelligent Control of general Nonlinear Discrete-Time Dynamical Systems: An analysis on the scalar case,” Int’l. Conf. on Data Science and Applications (ICDSA), 2021. (Kolkata, India)
- A. S. K. Sreeram, U. Ravishankar, N. R. Sripada, and B. Mamidgi, “Investigating the potential of MFCCs in

classifying respiratory diseases,” IEEE Intl. Symp. on Health and Medical Informatics, Management and Security (HMiMS), pp. 245 – 251, 2020. (Paris, France)

- U. Ravishankar, M. Manic and C. Rieger, “A micro-grid simulator tool (SGridSim) using Effective Node-to-Node Complex Impedance (EN2NCI) models,” IEEE Intl. Symp. On Resilient Control Systems (ISRCs), pp. 114 – 119, 2013. (San Francisco, CA – USA)
- U. Ravishankar and M. Manic, “Iterative Learning Heuristic Dynamic Programming (ILHDP) design of a Steam Power Plant Controller,” IEEE Industrial Electronics Conf. (IECON), pp. 1362 – 1367, 2012. (Montreal, Canada)
- U. Ravishankar and M. Manic, “The Adaptive Critic Learning Agent (ACLA) algorithm: Towards problem independent neural network based optimizers,” IEEE World Congress on Computational Intelligence (WCCI), pp. 1 – 7, 2012. (Brisbane, Australia)
- U. Ravishankar and M. Manic, “A Direct Utility Adaptive Critic (DUAC) algorithm for power plant load management,” IEEE Intl. Symp. on Industrial Electronics (ISIE), pp. 786 – 791, 2012. (Hangzhou, China)
- U. Ravishankar and M. Manic, “A hardware suitable Integrated Neural System for Autonomous Vehicles - Road Structuring and Path Tracking,” IEEE Intl. Joint Conf. on Neural Networks (IJCNN), pp. 2024 – 2030, 2011. (San Jose, CA – USA)
- J. Joshi, J. Zhang, C. Wang, C-C Hsu, A. C. Parker, C. Zhou and U. Ravishankar, “A Biomimetic Fabricated Carbon Nanotube Synapse for Prosthetic Applications,” IEEE Life Science Systems and Applications Workshop (LiSSA), pp. 139 – 142, 2011. (Bethesda, MD – USA)